CS F364: Design & Analysis of Algorithms

Quiz 6 — June 16, 2026

Coverage: Greedy Algorithms

> Note
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> **Instructions:**
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> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question: Interval Scheduling — Multiple Rooms

You are scheduling the following intervals in a venue with multiple identical rooms. Every interval must be scheduled in some room, and no two intervals in the same room may overlap. You want to use the **minimum number of rooms**.

Interval	I_1	I_2	I_3	I_4	I_5	I_6
Start	1	2	3	5	6	7
Finish	5	4	6	7	8	9

Describe the greedy algorithm for this problem (sorting rule and assignment rule). Run it step-by-step on the given intervals, showing which room each interval is assigned to. State the total number of rooms needed and explain why it is optimal.

Solution

Greedy algorithm: Sort intervals by start time (earliest first). For each interval in order, assign it to any existing room whose last scheduled interval has finished before this interval starts; if no such room exists, open a new room.

Step-by-step trace:

Step	Interval	Room	Rooms and their finish times
1	I_1 [1,5]	Room 1	R1: 5
2	I_2 [2,4]	Room 2	R1: 5, R2: 4
3	I_3 [3,6]	Room 3	R1: 5, R2: 4, R3: 6
4	I_4 [5,7]	Room 1	R1: 7, R2: 4, R3: 6
5	I_5 [6,8]	Room 2	R1: 7, R2: 8, R3: 6
6	I_6 [7,9]	Room 3	R1: 7, R2: 8, R3: 9

Total rooms needed: 3

Why optimal: At time interval [3,4], three intervals (I_1, I_2, I_3) are all active simultaneously. Any single room can hold at most one interval at a time, so at least 3 rooms are necessary. The greedy algorithm achieves exactly this lower bound, hence it is optimal.